Manuel Leite Pinho

Chemical Engineer | Frankfurt am Main, Germany

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Summary

I am a chemical engineer with 4 years' experience across process engineering, digital transformation, and research. Currently working as a process engineer supporting decarbonisation technologies. I'm challenge and optimisation oriented.

Work Experience

Process Engineer | Air Liquide | Dec 2024 – Present | Frankfurt am Main, Germany

- Developing dynamic simulations with Aspen HYSYS Dynamics.
- Creating a PFD generator tool using draw.io and leading internal training sessions.
- Designing simplified P&IDs for clients/suppliers.
- Simulating in Aspen Plus the CO2 to Methanol process.
- Executing dynamic plots with Python from raw plant data for a Root Cause Analysis.

Business Analyst / Project Manager | Galp | Jul 2023 – Nov 2024 | Lisbon, Portugal

- Managed technology and digitalisation projects across Oil & Gas extraction and refining, including setting up a drilling control room.
- Executed end-to-end process analysis and optimisation using SAP Signavio to identify and streamline inefficiencies.

Project Manager | Galp | May 2022 – Jun 2023 | Lisbon, Portugal

- Project management: SAP S/4HANA support and implementation in different geographies (Portugal, Spain, and Cabo Verde).
- Set up an automated tool in Power BI and Excel to internally assist ticketing classification.

Research Engineer | CERENA - Centro de Recursos Naturais e Ambiente | Feb 2022 – May 2022 | Lisbon, Portugal

- Modelled and controlled a biodiesel washing column with gPROMS ModelBuilder, MATLAB, and Simulink. Post-thesis work.

Production and Manufacturing Intern | The Navigator Company | Jul 2019 – Aug 2019 | Vila Velha de Ródão, Portugal

- Calibrated an industrial quality assurance scanning machine.

Education

Integrated Master in Chemical Engineering | Instituto Superior Técnico | Sep 2015 – Nov 2021 | Lisbon, Portugal

- Thesis: Dynamic Simulation of Biodiesel Washing Column.
- Optional subjects attended: process control, industrial chemistry and process optimisation.

International Summer School | Beihang University | Jul 2018 – Aug 2018 | Beijing, China

- Thermodynamics, Refrigeration technology, and Chinese culture.

Skills

- **Process Engineering:** Process Simulation, PFD, P&ID, Plant Layout, Scientific Modelling, Technical Drawing Software
- **General:** Microsoft Office, SAP S/4HANA, Microsoft Power BI, Python

Languages

Portuguese (Native) · **English** (C1) · **Spanish** (B1) · **French** (A1, currently studying)

Publications

- Rigorous Dynamic Modeling and Simulation of an Industrial Liquid–Liquid Extraction Packed Column for Biodiesel Washing | Ind. Eng. Chem. Res. (ACS), 2023 | First author
- Control of an Industrial Packed Extraction Column for Biodiesel Washing | ESCAPE-32 (Elsevier), 2022 | First author